EXCELLENCE

CLASS 11 - 2026-27

Physics | Motion 04

Motion in a Straight Line

Velocity as a vector · v–t graphs · when averages fail

By - Excellence Programme

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What is new today

01 Instantaneous velocity, and how it differs from speed

02 What a graph's slope and area actually give

03 v–t: a kink is not a U-turn

04 v = 0 with no reversal

05 Average velocity along the chord in 2-D

06 One nested average, and one impossible average

Left in Motion 03

Average speed, average velocity, the time-mix formula, the distance-mix formula, 4v/3, the 40 km/h round trip, and the three-thirds 16.4 m/s problem stay in lecture 03. We will not calculate them again.

Today the new job is direction: when velocity changes, what a v–t graph is allowed to say, and when an average cannot exist.

Instantaneous velocity

Average velocity looks at the whole trip. Instantaneous velocity is the velocity at one moment. In one dimension it is the slope of the x–t graph at that instant.

$$v = \frac{dx}{dt} \quad \cdot \quad \text{instantaneous speed} = |v|$$

- Differentiate x(t) when you want the instant, not the average.
- If the instantaneous velocity stays constant, the average velocity equals that same vector.

Speed fixed does not fix velocity

Speed is only the size. Velocity also carries direction. A particle can keep the same speed and still change its velocity by turning.

constant velocity → constant speed

constant speed → velocity may still change

variable speed → velocity is variable

Uniform circular motion is the clean picture: speed stays v, velocity keeps turning.

Rough work / extra work

|average velocity| cannot beat average speed

Distance is at least the straight shift, so the speed average is at least the size of the velocity average.

$$|\text{average velocity}| \leq \text{average speed}$$

They are equal only in one dimension when the object never reverses. Any turn, or any detour, makes the speed average strictly larger.

Zero links — speed and velocity

If this is zero	Then the other
average speed = 0	average velocity must be 0
average velocity = 0	average speed may still be > 0
average velocity ≠ 0	average speed must be > 0

A closed loop has average velocity zero and average speed greater than zero.

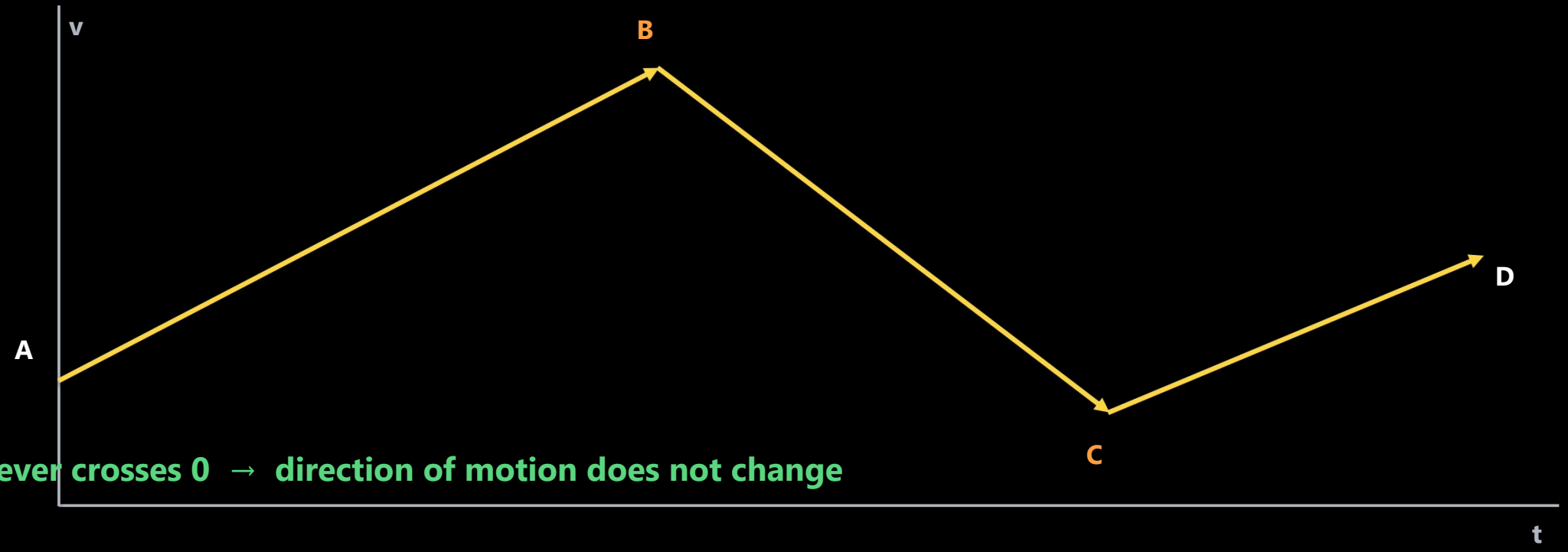
Four readings from one graph

Before a new graph, name the axes. Then only four things are available.

- The y-reading is the quantity on the vertical axis at that instant.
- The x-reading is the quantity on the horizontal axis.
- Slope = Δy / Δx. On an x–t graph the slope is velocity.
- Area under a v–t graph equals displacement (watch the sign below the axis).

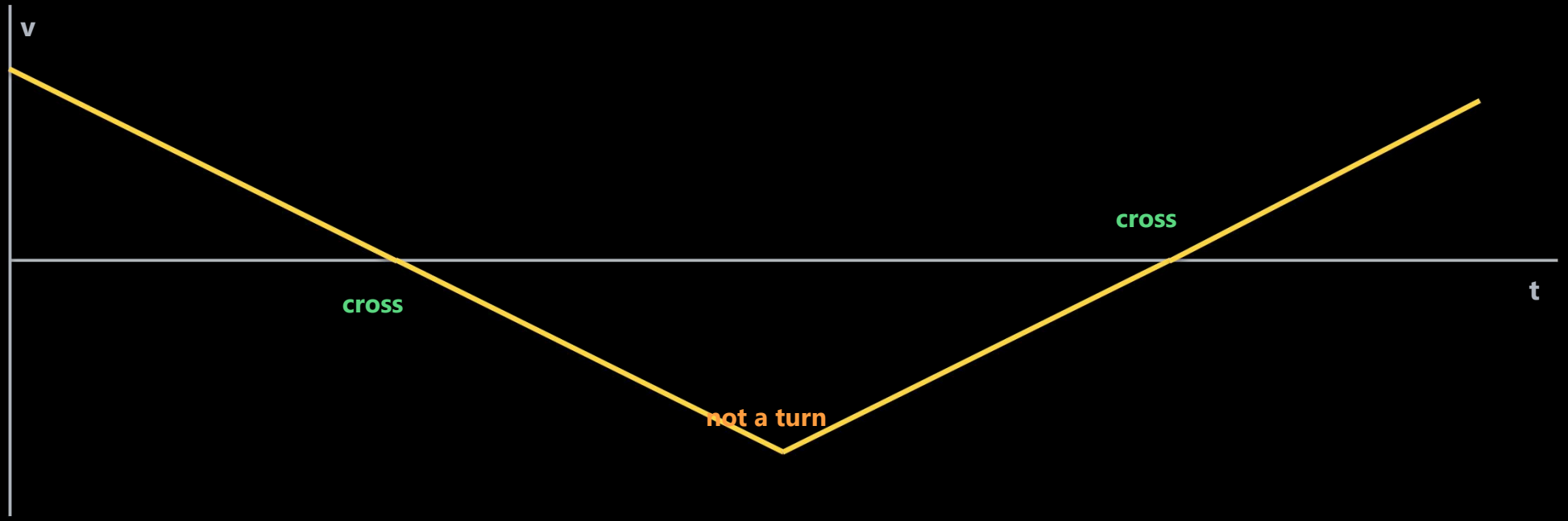
A kink on v–t is not a U-turn

Direction of motion changes only when velocity crosses zero. A sharp peak or a sharp valley, while v stays positive, is only a change of acceleration.



Where a real U-turn sits

On a v–t graph, mark only the crossings of the time axis. Those are the instants when velocity changes sign. A maximum or a minimum of speed, away from the axis, is not a turn-around.



Rough work / extra work

v = 0, and still no reversal

$x = t^3/3 - 2t^2 + 4t$. Find distance and displacement in the first 3 s.

$v = t^2 - 4t + 4 = (t - 2)^2$

v = 0 at t = 2 s, but $(t - 2)^2$ never becomes negative.

x(0) = 0 , x(3) = 3. No split of the path.

distance = 3 m · displacement = 3 m

A root of v = 0 is a U-turn only when v changes sign.

Read the start from $x = \alpha t^2 + \beta t - \gamma$

$x(0) = -\gamma$ → the object starts at $x = -\gamma$

$v = 2\alpha t + \beta$ → $v(0) = \beta$

$v(2\ s) = 4\alpha + \beta$

α, β and γ are constants given in the question. Do not treat them as variables you must eliminate.

Average speed from $v = 3t^2$

Speed at time t is $3t^2$ (already positive). Average speed over the first 2 s is the area under the speed–time curve, divided by 2 s.

area = ∫ from 0 to 2 of $3t^2$ dt = [t^3] from 0 to 2 = 8

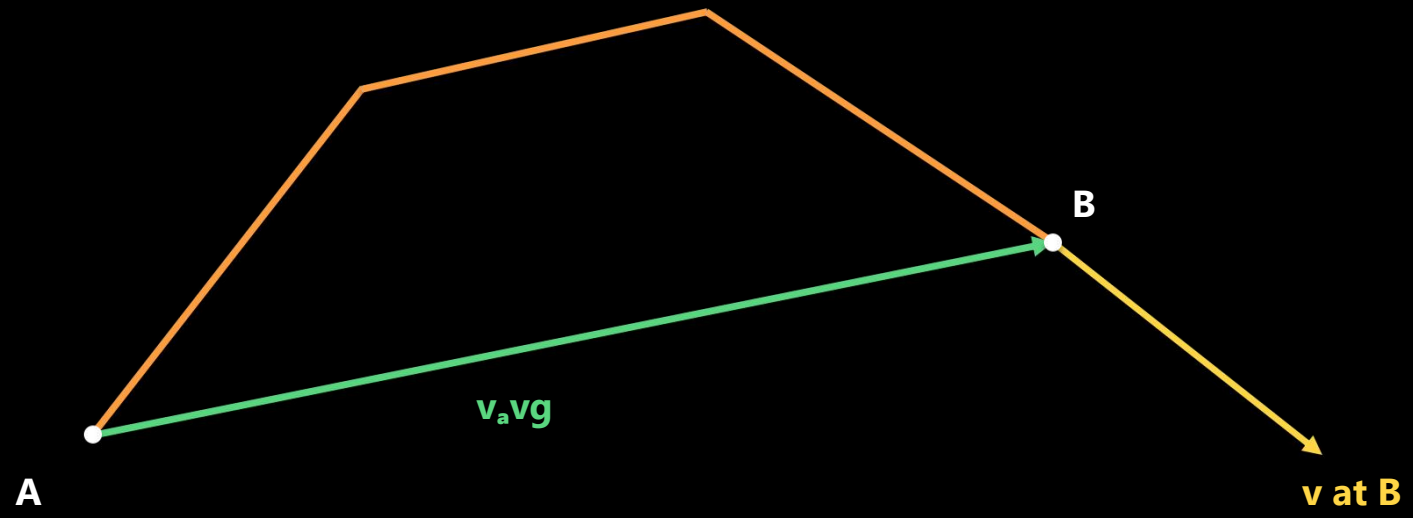
average speed = 8 / 2 = 4 m/s

Do not average v(0) and v(2) by adding and dividing by 2. The speed is not linear.

FIGURE

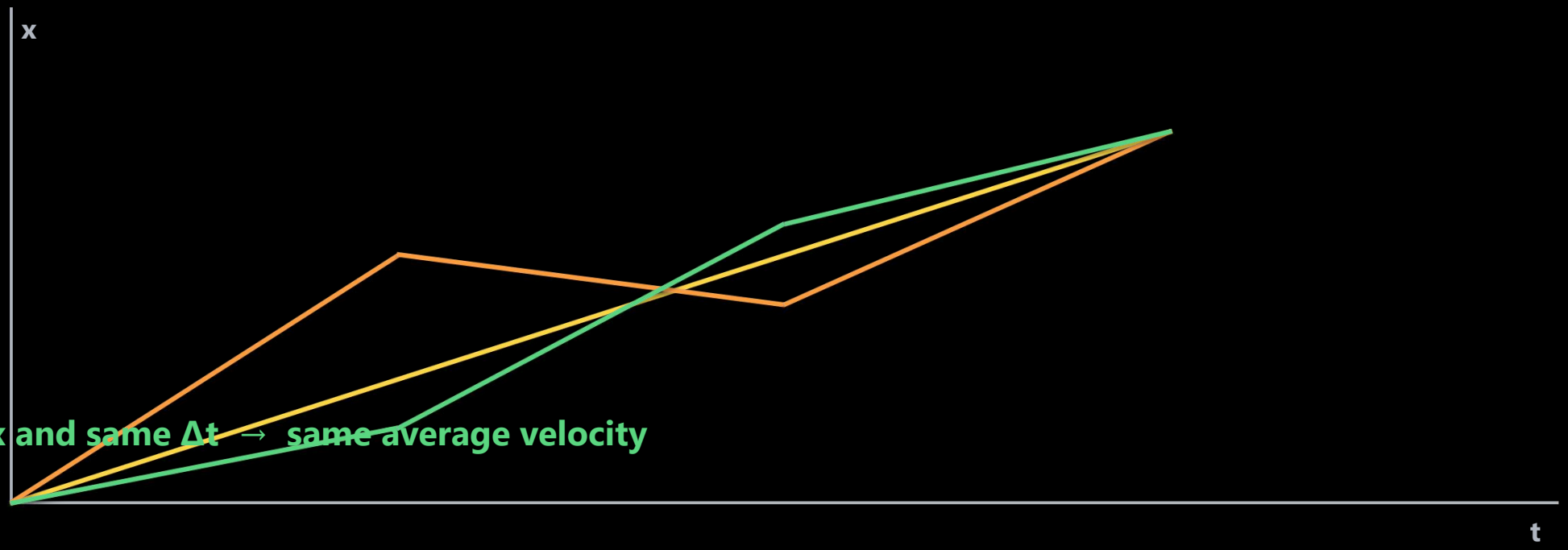
Chord versus tangent

On a curved path, average velocity points along the straight displacement from the start to the end. Instantaneous velocity points along the tangent at that moment.



Same end-points, same average velocity

Average velocity uses only the displacement and the time taken. Two different x–t curves that start together and finish together have the same average velocity, even if one is straight and one bends.



Rough work / extra work

Average velocity in a plane

At t = 0 the position is (2, 3) m. At t = 5 s the position is (13, 14) m. A point recorded at t = 2 s is not needed.

$$\Delta \vec{r} = (13 - 2)\hat{i} + (14 - 3)\hat{j} = 11\hat{i} + 11\hat{j}$$

$$\vec{v}_{avg} = \Delta \vec{r} / 5 = (11/5)\hat{i} + (11/5)\hat{j}$$

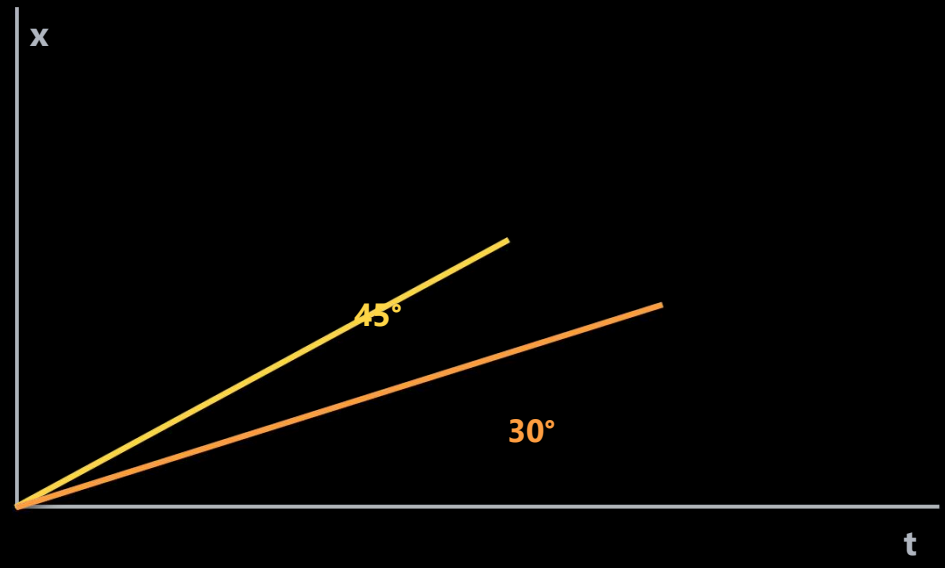
$$\vec{v}_{avg} = (11/5)(\hat{i} + \hat{j}) \text{ m/s}$$

Slope ratio on x–t

Two straight x–t lines leave the origin. Line A makes 45° with the time axis. Line B makes 30° with the time axis. Velocity is the slope.

vA / vB = tan 45° / tan 30°

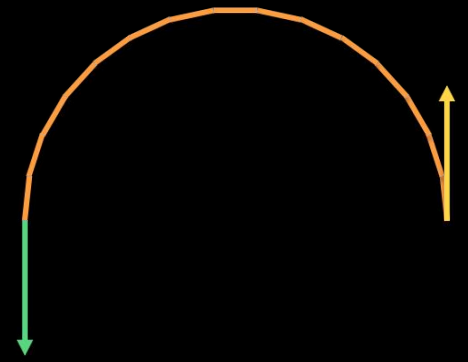
= 1 / (1/√3) = √3



MCQ — half a circle at steady speed

Uniform speed v on a circle. After half a circle, the change in velocity and the change in speed are

1. zero , zero
2. 2v , zero
3. 2v , 2v
4. zero , 2v



Solution: MCQ — half a circle at steady speed

Idea: the two velocity arrows have the same size and opposite direction, so their difference has size 2v. Speed never changed.

Answer: 2v , zero

A half that is itself an average

First half of the distance at 5 m/s. The second half is covered in two equal times, at 15 m/s and at 25 m/s.

On the second half, × are equal: (15 + 25) / 2 = 20 m/s

Whole trip: equal distances at 5 m/s and 20 m/s

average = 2 × 5 × 20 / (5 + 20) = 8 m/s

8 m/s

Rough work / extra work

An average that cannot exist

Equal distances. First half at 100 m/s. Someone claims the average for the whole trip is 200 m/s. Find the speed on the second half.

$2 \times 100 \times v / (100 + v) = 200$

$100\ v = 100(100 + v)$

$0 = 100$ → no real speed v

Average speed on equal distances always lies between the two speeds. 200 cannot sit above both if one of them is 100.

Only today’s lines

- v = dx/dt. Speed at that instant is |v|.
- |average velocity| ≤ average speed.
- Constant speed can still have a changing velocity.
- On v–t, direction changes only where the graph crosses zero.
- v = 0 with no sign change is not a U-turn.
- Average velocity follows the displacement chord, not the bend of the path.
- Area under a speed–time curve, divided by time, is average speed.

Thank you

Excellence · Motion 04 · Velocity · v–t · averages that are new